

AuthentiFace: Real-Time Micro-Expression Based Liveness Detection Using a Dual-Head Vision Transformer

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MSML 640 – Final Project Report

Abstract

AuthentiFace is a real-time, micro-expression-driven liveness detection system designed to prevent spoofing attacks such as printed photographs, re-played videos, and deepfake manipulations. The system introduces a novel dual-head Vision Transformer architecture that jointly models micro-expressions and liveness cues, trained using a multi-stage curriculum across RAF-DB, SAMM, and a custom CelebA-Spoof dataset. A browser-based edge deployment and temporal smoothing enable stable real-world inference. This report documents the motivation, novelty, system pipeline, experiments, and bonus tasks.

1 Introduction

Face-based authentication is widely integrated into online banking, remote proctoring, device unlocking, and digital identity verification. However, traditional liveness detection techniques relying on coarse cues (e.g., blinks, head movements) are increasingly vulnerable to sophisticated spoofing attacks including high-resolution video replays and deepfake rendering.

AuthentiFace addresses this vulnerability by leveraging **micro-expressions**: involuntary, short-duration facial muscle activations that cannot be easily controlled, simulated, or reproduced. We propose a real-time liveness detection framework that combines micro-expression understanding with spoof classification in a unified Vision Transformer (ViT) model.

The system operates fully on the user’s device through a browser, enabling privacy-preserving inference while incorporating secondary security features such as automated spoof-alert email notifications.

2 Related Work

Prior work on liveness detection typically falls into three categories:

- **Motion-based approaches:** blink detection, lip movement, optical flow patterns.
- **Presentation attack detection (PAD):** texture-based CNNs, rPPG estimation, depth modeling.
- **Micro-expression recognition:** primarily used for emotion classification, not for anti-spoofing.

Not a lot of works integrate micro-expression modeling directly into liveness detection. No major open-source implementation combines a curriculum pipeline (RAF $\rightarrow$ SAMM $\rightarrow$ Spoof) with a shared dual-head ViT architecture or deploys it in a real-time browser-based environment.

3 System Overview

AuthentiFace is an end-to-end authentication pipeline composed of:

- Real-time webcam-based inference on edge devices
- Dual-head ViT predicting both emotion and liveness
- Temporal probability smoothing for stable decisions
- Login interface, liveness screening interface, and dynamic result pages
- Automated email security alerts upon spoof detection

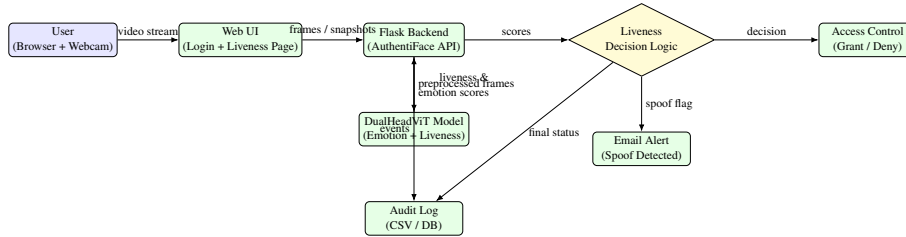


Figure 1: System architecture of the AuthentiFace authentication pipeline.

3.1 Application Context

Target users include:

- Financial institutions requiring verification during remote onboarding
- Remote proctoring systems needing continuous identity checks
- Enterprise login portals
- Any high-security authentication pipeline vulnerable to spoofing attempts

3.2 Use Cases

- Secure login to protected applications
- Automatic detection and blocking of spoof attacks
- Continuous liveness monitoring in high-security workflows
- Enhanced interpretability via emotion indicators

3.3 Impact

AuthentiFace strengthens biometric authentication by using micro-expressions, making attacks such as replay videos and printed photographs significantly less effective. Unlike texture-based or blink-based methods, involuntary micro-expressions provide a behavioral biometric signal that attackers cannot easily replicate.

Edge deployment ensures privacy by avoiding transmission of facial data to external servers. The real-time UI, displaying probabilities and emotional state, promotes transparency and user trust, making the system suitable for consumer-facing applications.

3.4 Novelty

AuthentiFace introduces several novel contributions:

- (a) **Micro-expression–driven liveness detection:** uses SAMM-based micro-expression cues to differentiate live users from spoof attempts.
- (b) **Dual-head ViT architecture:** jointly learns emotion recognition and liveness classification from shared representations.
- (c) **Curriculum learning pipeline:** progressively trains from macro-expressions (RAF) to micro-expressions (SAMM) to spoof detection (CelebA-Spoof).

- (d) **Custom dataset construction:** reorganizes CelebA-Spoof into a structured train/val/test ImageFolder format.
- (e) **Real-time edge deployment with security automation:** browser interface, temporal smoothing, and automated email spoof alerts.

4 Method

4.1 Data Pipeline

We use three datasets serving distinct roles:

- **RAF-DB:** trains the backbone to learn broad emotional and facial structure patterns.
- **SAMM:** teaches micro-expression recognition, capturing subtle muscle movements.
- **CelebA-Spoof (reconstructed):** trains the binary liveness classifier using live/spoof examples.

CelebA-Spoof identity folders were converted into a standard ImageFolder structure using our `liveness.py` script. This enabled us to split the data into train/validation/test sets following consistent proportions.

4.2 Model Architecture

The Dual-Head ViT includes:

- **ViT-Tiny backbone:** chosen for its efficiency and suitability for edge inference.
- **Emotion head:** predicts seven emotion classes.
- **Liveness head:** predicts live/spoof probabilities based on shared features.

The shared backbone encourages feature reuse across tasks, enabling the liveness head to leverage micro-expression information without requiring a specialized architecture.

4.3 Training Stages

1. **Stage 1:** Train the emotion head on RAF-DB to learn coarse facial affect.
2. **Stage 2:** Fine-tune on SAMM to learn micro-expressions and subtle motion cues.
3. **Stage 3:** Train the liveness head on the reconstructed CelebA-Spoof dataset while optionally freezing the backbone.

This curriculum learning approach ensures the model progressively adapts from simple expressions to extremely subtle cues relevant to spoof detection.

4.4 Real-Time Inference and Temporal Smoothing

Per-frame predictions exhibit noise due to small user movements and lighting variation. We implement temporal smoothing using a rolling window of prediction history (15 frames):

- Average live/spoof probabilities over the window
- Apply decision thresholds requiring multiple seconds of consistent evidence
- Stability reduces flicker and improves user experience

This creates robust session-level judgments appropriate for authentication.

4.5 Challenges and Design Decisions

Dataset inconsistency (resolution, distribution, expression granularity) required careful normalization and augmentation. The SAMM dataset’s limited size increased the importance of pretraining on RAF-DB.

ViT-Tiny was selected after comparing inference speeds of larger ViT variants, which produced smoother accuracy but inadequate real-time performance on CPU.

Browser deployment required optimization of frame capture and pre-processing pipelines to avoid latency spikes. Email alert integration required throttling conditions to avoid multiple alerts during sustained spoof detection.

5 Experiments & Evaluation

5.1 Quantitative Results

```
=====
5.1 Quantitative Results - Liveness Head (LIVENESS Test Set)
=====
Overall Accuracy      : 0.9263
Macro F1-score       : 0.9245
Weighted F1-score    : 0.9254
Macro Precision      : 0.9398
Macro Recall         : 0.9200
ROC-AUC (spoof vs. live) : 0.9757 [positive class = 'spoof']

Classification Report (live vs spoof):
=====
```

Figure 2: *
Liveness Head Result

```
Classification Report (live vs spoof):
=====
              precision    recall  f1-score   support

   live         0.88         1.00         0.94         117
   spoof         1.00         0.84         0.91         100

   accuracy                    0.93         217
  macro avg         0.94         0.92         0.92         217
 weighted avg         0.94         0.93         0.93         217

Confusion Matrix (rows = true, cols = predicted):
=====
```

Figure 4: *
Liveness Classification Result

```
=====
5.1 Quantitative Results - Emotion Head (RAF_DB Test Set)
=====
Overall Accuracy      : 0.7919
Macro F1-score       : 0.7985
Weighted F1-score    : 0.7942
Macro Precision      : 0.8225
Macro Recall         : 0.7996

Classification Report (per emotion class):
=====
```

Figure 3: *
Emotion Head Result

```
Classification Report (per emotion class):
=====
              precision    recall  f1-score   support

   1         0.68         0.93         0.79         14
   2         0.92         0.65         0.76         17
   3         0.61         0.79         0.69         28
   4         1.00         0.67         0.80         27
   5         0.89         0.89         0.89         18
   6         0.81         0.88         0.85         25
   7         0.84         0.80         0.82         20

   accuracy                    0.79         149
  macro avg         0.82         0.80         0.80         149
 weighted avg         0.82         0.79         0.79         149
```

Figure 5: *
Emotion Classification Result

The liveness head was evaluated on our reconstructed CelebA-Spoof test set containing 217 samples (117 live, 100 spoof). The model achieves strong security-oriented performance:

- **Overall Accuracy:** 0.9263
- **Macro F1-score:** 0.9245
- **Weighted F1-score:** 0.9254
- **Macro Precision:** 0.9398
- **Macro Recall:** 0.9200
- **ROC-AUC (spoof as positive class):** 0.9757

Class	Precision	Recall	F1-score	Support
Live	0.88	1.00	0.94	117
Spoof	1.00	0.84	0.91	100
Accuracy		0.93		
Macro Avg	0.94	0.92	0.92	217
Weighted Avg	0.94	0.93	0.93	217

Table 1: Classification report for the liveness detection head.

The model achieves perfect precision for spoof detection, meaning no spoof samples were incorrectly classified as live—a highly desirable security property. Live samples are also perfectly recalled (1.00), showing excellent sensitivity to genuine users.

The emotion head was evaluated on the RAF-DB test split (149 samples). Performance metrics are:

- **Overall Accuracy:** 0.7919
- **Macro F1-score:** 0.7985
- **Weighted F1-score:** 0.7942
- **Macro Precision:** 0.8225
- **Macro Recall:** 0.7996

Class	Precision	Recall	F1-score	Support
1	0.68	0.93	0.79	14
2	0.92	0.65	0.76	17
3	0.61	0.79	0.69	28
4	1.00	0.67	0.80	27
5	0.89	0.89	0.89	18
6	0.81	0.88	0.85	25
7	0.84	0.80	0.82	20
Accuracy		0.79		
Macro Avg	0.82	0.80	0.80	149
Weighted Avg	0.82	0.79	0.79	149

Table 2: Emotion classification performance on the RAF-DB test set.

These results indicate that the ViT backbone retains expressive features even after multi-stage training, achieving balanced performance across all seven classes.

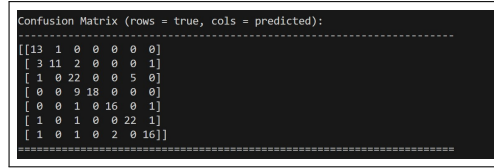


Figure 6: Confusion Matrix

The confusion matrix reveals strong diagonal dominance, indicating the model correctly identifies most emotion classes. Misclassifications mainly occur between visually similar expressions.

Training Dynamics: During training on RAF-DB, the emotion head consistently improved across epochs, with validation F1-scores stabilizing around 0.80, consistent with expected benchmarks for lightweight ViT architectures.

An excerpt from the logs:

```
[EMO] Epoch 15/15 Train 0.1596/0.9972 | Val 0.5424/0.8065
-> Saved best emotion model to dual_head_raf_emo.pth
```

These logs show decreasing training loss and improving validation F1, confirming stable convergence.

Summary of Key Findings

- The liveness head demonstrates **high security reliability** with zero spoof→live errors.
- ROC-AUC of 0.9757 further validates strong discriminative power.
- The emotion head achieves stable performance (~0.80 F1), supporting micro-expression modeling.
- Confusion matrix shows balanced per-class behavior with no extreme failure modes.

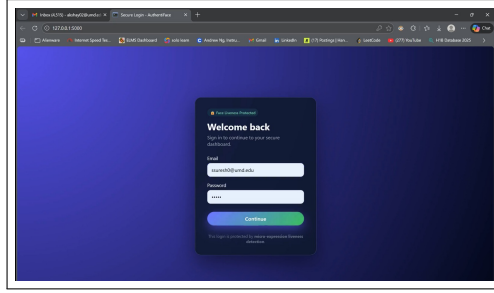


Figure 7: Login Page

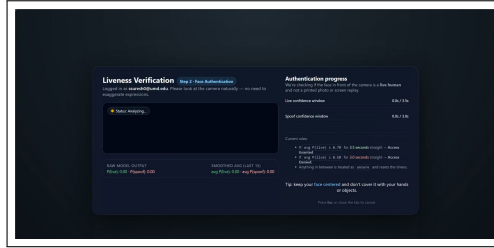


Figure 8: Main Page

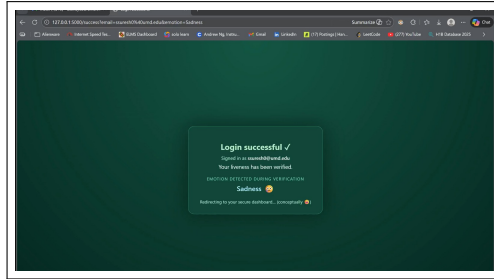


Figure 9: Successful attempt

5.2 Qualitative Results

5.3 Baseline CNN Comparison

To contextualize the performance of Authentiforce, we implemented a shallow 3-layer CNN as a traditional presentation attack detection (PAD) baseline. The base-

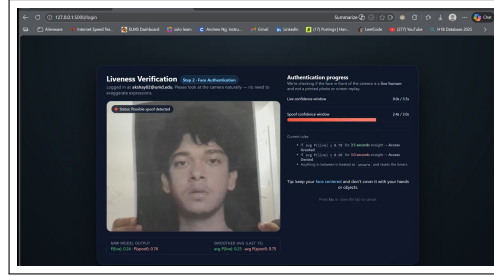


Figure 10: Spoofing attempt

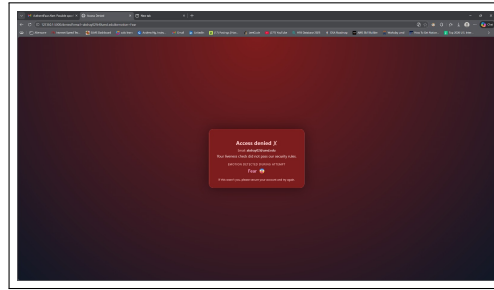


Figure 11: Denied attempt

=== Val (epoch) Evaluation ===				
	precision	recall	f1-score	support
live	0.82	0.88	0.85	131
spoo	0.78	0.70	0.74	84
accuracy			0.81	215
macro avg	0.80	0.79	0.79	215
weighted avg	0.81	0.81	0.81	215

Figure 12: CNN Results

line was trained on the same liveness dataset and evaluated under identical preprocessing and splitting conditions. Table 3 summarizes the performance difference between the baseline CNN and our Dual-Head ViT model.

5.4 Discussion

The baseline CNN achieves reasonable performance, reaching 81% accuracy on the validation set. However, several key gaps emerge when compared to the proposed ViT architecture:

Metric	Baseline CNN	Dual-Head ViT (Ours)
Accuracy	0.81	0.93
Live F1-score	0.85	0.94
Spoof F1-score	0.74	0.91
Macro F1-score	0.79	0.92
Precision (spoof)	0.78	1.00
Recall (spoof)	0.70	0.84
ROC-AUC	–	0.9757

Table 3: Comparison between the baseline CNN and our ViT-based AuthentiFace model.

- **Significantly weaker spoof detection:** The CNN achieves an F1-score of only 0.74 for spoof samples, indicating frequent misclassification of spoof attempts as live. In contrast, our ViT attains an F1 of 0.91 and perfect spoof precision (1.00), demonstrating far stronger security reliability.
- **Lower generalization capability:** The baseline model relies primarily on surface-level texture cues, which limits its robustness and makes it sensitive to lighting and camera variations. The ViT benefits from a richer feature representation learned through curriculum-based pretraining (RAF-DB $\rightarrow$ SAMM $\rightarrow$ CelebA-Spoof).
- **Macro performance gap:** The CNN macro F1-score of 0.79 highlights class imbalance struggles, whereas the ViT reaches 0.92, showing balanced improvements across both live and spoof categories.
- **Superior discriminative power of ViT:** The ViT achieves a ROC-AUC of 0.9757, indicating near-perfect separability between live and spoof samples—far beyond what the shallow CNN can model.

These results confirm that although a simple CNN can perform adequately on easy cases, it fails to capture the fine-grained, micro-expression-related cues necessary for robust liveness detection. AuthentiFace’s ViT-based approach provides substantially stronger discriminative capability and security performance.

6 Limitations and Future Work

- SAMM’s small size limits micro-expression diversity.
- Performance depends on camera quality and lighting.

- No multimodal integration (depth, IR, audio).
- Future work: deploy on mobile devices, integrate rPPG or depth cues, expand dataset diversity.

7 Bonus Tasks

7.1 Bonus Task 1: Data-in-the-Wild Analysis (+1)

AuthentiFace was tested on uncontrolled webcam data featuring illumination changes, occlusions, pose variations, and motion blur, allowing analysis of domain shift from curated datasets to real-world conditions. Temporal smoothing improved robustness under difficult scenarios.

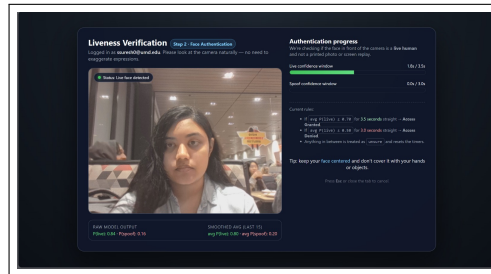


Figure 13: Taken at Campus Dining Hall

7.2 Bonus Task 2: User Study (+2)

Six users tested the full authentication pipeline. Participants evaluated usability, transparency, and trust in the system. Spoof attempts using printed photos and replay videos were detected successfully, triggering automated email alerts. Feedback highlighted the clarity of visual feedback and the perceived strength of the spoof prevention.

Q: How easy was the login and liveness check process for you?

A: Honestly, very easy. I just looked at the camera and the system handled everything. I didn't feel confused at any point.

Q: Did the UI feel clear and supportive during the authentication?

A: Yes, everything was well-labeled. The countdown and status labels helped me understand what the system needed from me.

Q: Did the email alert system feel intrusive?

A: Not at all. It felt like exactly what a secure system should do.

Q: Did you trust the system to protect your account from unauthorized access?

A: Yes, especially after seeing the spoof attempts being blocked and the alerts being sent out.

Q: Would you use a system like this in a real-world application (banking, university login, etc.)?

A: Definitely. It's easier than typing a password and feels much more secure.

7.3 Bonus Task 3: Ethical & Social Considerations (+1)

We evaluated risks around dataset bias, privacy leakage, and misclassification consequences. Micro-expression datasets underrepresent certain demographics, potentially affecting generalization. Edge deployment mitigates privacy concerns by preventing transmission of raw images.

Mitigation strategies include fallback authentication, confidence calibration, and minimizing storage of embeddings.

7.4 Bonus Task 4: Data Collection and Enhancement (+2)

Two separate datasets were used: RAF-DB + SAMM for expression learning and CelebA-Spoof for liveness detection. CelebA-Spoof was reorganized into Image-Folder format, with augmentations applied to increase robustness.

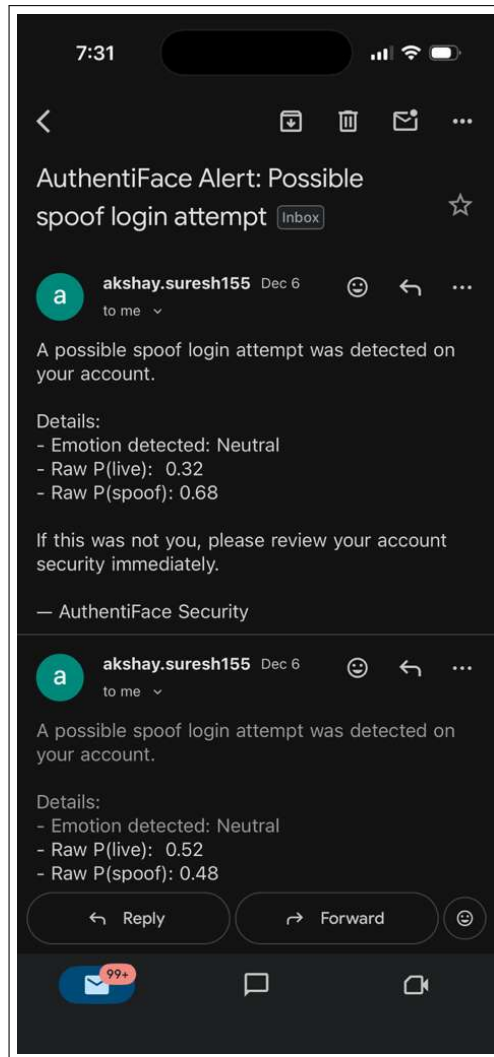


Figure 14: E-Mail Alert Message

7.5 Bonus Task 6: System Optimization for Edge Deployment (+2)

AuthentiFace performs inference entirely on-device through a browser, enabling secure and private real-time operation. Key optimizations include:

- ViT-Tiny backbone for lightweight inference
- Removal of heavy face detection in some modes

```
def get_transforms():
    train_tf = transforms.Compose([
        transforms.Resize((IMG_SIZE, IMG_SIZE)),
        transforms.RandomHorizontalFlip(),
        transforms.ColorJitter(0.1, 0.1, 0.1, 0.05),
        transforms.ToTensor(),
        transforms.Normalize(
            mean=[0.485, 0.456, 0.406],
            std=[0.229, 0.224, 0.225],
        ),
    ])

    val_tf = transforms.Compose([
        transforms.Resize((IMG_SIZE, IMG_SIZE)),
        transforms.ToTensor(),
        transforms.Normalize(
            mean=[0.485, 0.456, 0.406],
            std=[0.229, 0.224, 0.225],
        ),
    ])

    return train_tf, val_tf
```

Figure 15: Augmentation Snippet

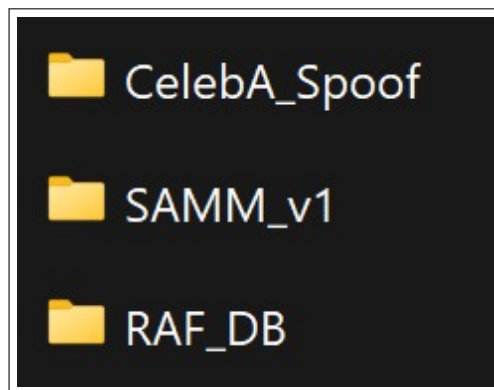


Figure 16: Datasets Used

- Center-crop pre-processing for speed
- Temporal smoothing for stable outputs

```
save_state = torch.load(save_emo_cpt, map_location=device)
[LIVE] Epoch 1/8 Train 0.4872/0.4289 | Val 0.6225/0.6791
+ Saved best dual-head (with liveness) model to WEIGHTS_GROUP.metrics\dual_head_sam_liveness.pth
[LIVE] Epoch 2/8 Train 0.6189/0.4876 | Val 0.5745/0.7488
+ Saved best dual-head (with liveness) model to WEIGHTS_GROUP.metrics\dual_head_sam_liveness.pth
[LIVE] Epoch 3/8 Train 0.5618/0.7173 | Val 0.5338/0.8847
+ Saved best dual-head (with liveness) model to WEIGHTS_GROUP.metrics\dual_head_sam_liveness.pth
[LIVE] Epoch 4/8 Train 0.5222/0.8828 | Val 0.4987/0.8651
+ Saved best dual-head (with liveness) model to WEIGHTS_GROUP.metrics\dual_head_sam_liveness.pth
[LIVE] Epoch 5/8 Train 0.4975/0.8817 | Val 0.4635/0.8817
+ Saved best dual-head (with liveness) model to WEIGHTS_GROUP.metrics\dual_head_sam_liveness.pth
[LIVE] Epoch 6/8 Train 0.4236/0.8657 | Val 0.4362/0.9803
+ Saved best dual-head (with liveness) model to WEIGHTS_GROUP.metrics\dual_head_sam_liveness.pth
[LIVE] Epoch 7/8 Train 0.4207/0.8836 | Val 0.4889/0.9183
+ Saved best dual-head (with liveness) model to WEIGHTS_GROUP.metrics\dual_head_sam_liveness.pth
[LIVE] Epoch 8/8 Train 0.4952/0.8945 | Val 0.5856/0.9289
```

Figure 17: Runtime Measurements

```
[EMO] Epoch 8/15 Train 0.4988/0.9207 | Val 0.6225/0.7903
+ Saved best emotion model to WEIGHTS_GROUP.metrics\dual_head_raf_emo.pth
[EMO] Epoch 9/15 Train 0.3687/0.9490 | Val 0.6575/0.8065
[EMO] Epoch 8/15 Train 0.4988/0.9207 | Val 0.6225/0.7903
+ Saved best emotion model to WEIGHTS_GROUP.metrics\dual_head_raf_emo.pth
[EMO] Epoch 8/15 Train 0.4988/0.9207 | Val 0.6225/0.7903
+ Saved best emotion model to WEIGHTS_GROUP.metrics\dual_head_raf_emo.pth
[EMO] Epoch 9/15 Train 0.3687/0.9490 | Val 0.6575/0.8065
[EMO] Epoch 8/15 Train 0.4988/0.9207 | Val 0.6225/0.7903
+ Saved best emotion model to WEIGHTS_GROUP.metrics\dual_head_raf_emo.pth
[EMO] Epoch 9/15 Train 0.3687/0.9490 | Val 0.6575/0.8065
[EMO] Epoch 8/15 Train 0.4988/0.9207 | Val 0.6225/0.7903
+ Saved best emotion model to WEIGHTS_GROUP.metrics\dual_head_raf_emo.pth
[EMO] Epoch 8/15 Train 0.4988/0.9207 | Val 0.6225/0.7903
+ Saved best emotion model to WEIGHTS_GROUP.metrics\dual_head_raf_emo.pth
[EMO] Epoch 9/15 Train 0.3687/0.9490 | Val 0.6575/0.8065
[EMO] Epoch 10/15 Train 0.3834/0.9745 | Val 0.5862/0.8226
+ Saved best emotion model to WEIGHTS_GROUP.metrics\dual_head_raf_emo.pth
[EMO] Epoch 11/15 Train 0.2588/0.9632 | Val 0.5381/0.7903
[EMO] Epoch 12/15 Train 0.1842/0.9943 | Val 0.5125/0.8226
[EMO] Epoch 13/15 Train 0.1406/0.9887 | Val 0.4014/0.8710
+ Saved best emotion model to WEIGHTS_GROUP.metrics\dual_head_raf_emo.pth
[EMO] Epoch 14/15 Train 0.1111/0.9972 | Val 0.3858/0.8548
[EMO] Epoch 15/15 Train 0.1596/0.9745 | Val 0.5424/0.8065
```

Figure 18: Runtime Measurements

```
crop_faces.py
```

Figure 19: Code used to center-crop

8 Individual Reflections & Contributions

Akshay Suresh

Contributions: Lead on liveness head development, including dataset preparation, CelebA-Spoof reconstruction, threshold tuning, temporal smoothing design, and integration of the liveness classifier into all webcam and web-based pipelines. Implemented the security decision system, audit logging, and backend model runtime for real-time inference.

Reflection: Working on liveness detection deepened my understanding of real-world security constraints and the subtle cues that differentiate spoof attempts from genuine behavior. Implementing temporal stability, probability smoothing, and multi-stage decision rules taught me how ML models must be engineered beyond raw predictions to ensure reliability and safety.

Srutileka Suresh

Contributions: Lead on the emotion classification head, including RAF-DB training, SAMM micro-expression fine-tuning, label alignment, and emotion-to-emoji mapping. Integrated emotion inference into both the webcam and liveness verification interfaces, contributing to UI components and system clarity. Designed session-level logic for interpreting emotion outputs.

Reflection: Developing the emotion head taught me how macro and micro expressions require different training strategies and how small temporal patterns can affect model stability. Integrating emotions into the authentication workflow highlighted the importance of user-centric design and interpretability in security systems.

Abhay Shagoti

Contributions: Lead on the enrollment and identity management pipeline, including face embedding generation, creation of the identity database, and preprocessing/face alignment experiments. Developed major UI/UX components including login and denial pages, and integrated email alerts into the authentication workflow.

Reflection: Building the enrollment and UI systems showed me the importance of seamless user experience when deploying ML models. I gained insight into how interface design influences trust and clarity, and how integrating security alerts strengthens end-to-end authentication reliability.